SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 2 (TOPIC 2)

Theme: MICROCOMPUTER COMPONENTS, SUBSYSTEMS & ARCHITECTURAL COMPARISONS

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 2.1 (Devices & Logical Configurations: CPU, Memory, Bus, I/O, Storage), Subtopic 2.2 (Comparative Aspects)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 4 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Show all intermediate formulas, binary conversions, architectural diagrams, command syntax, and arithmetic calculations clearly.
4. Where design questions require software configurations, provide exact command line syntax (Bash/DOS) or functional formulas (Excel) as specified.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL ARCHITECTURAL CONCEPTS [25 MARKS]

Question 1 [5 Marks]: Detail the functional role of the four internal CPU components: (a) Arithmetic Logic Unit (ALU), (b) Control Unit (CU), (c) Program Counter (PC), and (d) Instruction Register (IR). Describe how they collaborate during instruction decoding.

Question 2 [5 Marks]: Differentiate between Static RAM (SRAM) and Dynamic RAM (DRAM) at the transistor/circuit level. Explain why SRAM is utilized exclusively for processor cache hierarchies (L1, L2, L3) while DRAM is utilized for main system memory.

Question 3 [5 Marks]: Explain the mechanical architecture of a Hard Disk Drive (HDD). Define the components that govern total average data access time: Seek Time, Rotational Latency, and Transfer Time. How does a Solid-State Drive (SSD) eliminate two of these components?

Question 4 [5 Marks]: Define the three functional sub-buses comprising the system bus: (a) Data Bus, (b) Address Bus, and (c) Control Bus. Specify which of these buses are unidirectional and which are bidirectional, providing rigorous architectural reasons.

Question 5 [5 Marks]: Compare CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) design philosophies. Explain how modern microprocessors (such as Intel Core x86-64 and Apple M-Series ARM) incorporate elements of both philosophies.

SECTION B: QUANTITATIVE HARDWARE & BUS CALCULATIONS [35 MARKS]

Question 6 [10 Marks]: A magnetic enterprise hard disk drive spins at 7,200 revolutions per minute (RPM). It has an average seek time of 8.5 ms and a sustained track transfer rate of 180 MB/s.
(a) Calculate the rotational latency for one complete disk revolution in milliseconds. [3 Marks]
(b) Compute the average rotational latency (defined as half of one revolution) in milliseconds. [2 Marks]
(c) Determine the total average time required to locate and read a single 64 KB sequential data block from the disk. [5 Marks]

Question 7 [10 Marks]: Compute the theoretical peak bandwidth for the following microcomputer expansion buses:
(a) A legacy 32-bit PCI bus operating at 33.33 MHz. Express your answer in Megabytes per second (MB/s). [3 Marks]
(b) A modern PCI Express 4.0 expansion slot with 16 physical lanes (x16). (Note: PCIe 4.0 transfers at 16.0 GT/s per lane using 128b/130b encoding). Express bandwidth in both Gigatransfers per second and Gigabytes per second (GB/s). [4 Marks]
(c) A dual-channel DDR4 memory subsystem operating at a clock rate of 3,200 MT/s (MegaTransfers/sec) with a 64-bit bus per channel. Calculate peak bandwidth in GB/s. [3 Marks]

Question 8 [8 Marks]: A microcomputer processor features a three-level cache hierarchy:
- L1 Cache: Access latency = 1.2 ns, Hit rate = 92%
- L2 Cache: Access latency = 4.5 ns, Hit rate = 97% (local hit rate of misses from L1)
- L3 Cache: Access latency = 12.0 ns, Hit rate = 90% (local hit rate of misses from L2)
- Main DRAM Memory: Access latency = 65.0 ns
Calculate the Effective Memory Access Time (EMAT) for this microcomputer memory hierarchy. [8 Marks]

Question 9 [7 Marks]: Solid-State Storage Engineering:
(a) Contrast Single-Level Cell (SLC), Multi-Level Cell (MLC), Triple-Level Cell (TLC), and Quad-Level Cell (QLC) NAND flash memory in terms of bit storage density, access latency, and write endurance (P/E cycles). [4 Marks]
(b) Explain why the operating system TRIM command and wear-leveling algorithms are essential for maintaining SSD write performance and preventing premature drive failure. [3 Marks]

SECTION C: SYSTEM COMPARISONS & APPLIED INTEGRATION [40 MARKS]

Question 10 [15 Marks]: Workstations versus Standard Commercial Microcomputers:
The Department of Soil and Geological Sciences at SUA requires a high-performance computer for running 3D digital elevation models (DEM) and hyperspectral soil GIS simulations.
(a) Contrast an engineering workstation with an ordinary consumer desktop PC across: Processor architecture (e.g., AMD Threadripper/Intel Xeon vs Core i5), Memory architecture (ECC vs non-ECC), GPU capabilities (Nvidia RTX Ada Generation vs consumer GeForce), and Storage interfaces. [7 Marks]
(b) What is Error-Correcting Code (ECC) memory? Explain the physical Hamming code mechanism that enables single-bit error correction and double-bit error detection (SEC-DED). Why is this critical for long-running scientific simulations? [5 Marks]
(c) Provide a cost-benefit recommendation for the department head justifying whether to purchase 1 workstation ($4,500) or 3 standard desktop PCs ($1,500 each). [3 Marks]

Question 11 [15 Marks]: Comparative Analysis: Microcomputers vs Mainframes and Minicomputers:
(a) Construct an exhaustive comparison matrix evaluating Microcomputers, Minicomputers (Midrange systems), and Mainframes across: Typical user concurrency, I/O processor subsystem architecture, operating system complexity, physical footprints, and fault-tolerant hardware redundancy. [8 Marks]
(b) Trace the historical paradigm shift that led to the decline of traditional minicomputers (e.g., DEC PDP and VAX) in favor of networked microcomputer client-server architectures in university and enterprise environments. [4 Marks]
(c) Explain the concept of 'Symmetric Multiprocessing' (SMP) and how modern multi-socket microcomputer servers achieve processing capabilities that rival historic enterprise mainframes. [3 Marks]

Question 12 [10 Marks]: Peripheral Interface & Agricultural Sensor Integration:
An engineering student at SUA is designing an automated greenhouse control unit based on an embedded microcomputer (Raspberry Pi 4B) connected to soil moisture sensors, temperature probes, and irrigation solenoid relays.
(a) Differentiate between Analog-to-Digital Conversion (ADC) and Digital I/O (GPIO). Why does the microcomputer require an external ADC (such as the ADS1115) to read analog resistive soil sensors? [4 Marks]
(b) Contrast the I2C, SPI, and UART serial communication protocols in terms of wiring complexity, maximum data rate, and multi-device addressing. Recommend the most appropriate protocol for connecting 6 identical digital environmental sensors. [6 Marks]